

A05 Contribution Journal

Analyzing Arrival Through the Lens of NLP

Team: DataProxy

Course: ITAI 2376

Movie: Arrival (2016)

Journal Owner: Elijah Raines

Project Overview

For A05, our team is assigned to examine how *Arrival* (2016) presents communication with an unfamiliar language and connect those scenes to natural language processing. The planned report will cover the movie's central communication problem, ambiguity and context, language-learning methods, three NLP approaches, modern tools, and a final reflection. This journal records the work each member is responsible for contributing so the group can combine the parts into one clear report.

Team and Assigned Roles

Team Member	Assigned Role
Jonah Joseph	Movie Overview / Introduction
Fifehanmi Ogedengbe	NLP Challenges in the Movie
Justin Davis	How They Learn the Alien Language / Communication Methods
Brandon Matias	NLP Approaches (rule-based, statistical, deep NLP)
Samantha Mireles	Real-World NLP Tools and Examples
Elijah Raines	Reflection, final lesson, references, and final assembly

Assigned Contributions

Jonah Joseph

Assigned role: *Movie Overview / Introduction*

Jonah is responsible for opening the report with the background needed for the team's analysis. His planned section will identify Louise Banks as the linguist asked to communicate with the heptapods, explain the central problem of interpreting a language with no shared starting point, and give only the plot context needed to understand why accurate interpretation matters.

Fifehanmi Ogedengbe

Assigned role: *NLP Challenges in the Movie*

Fifehanmi is assigned to explain the language-processing challenges shown in the movie. Her section will address ambiguity, missing context, cultural differences, and the risk of making assumptions too quickly. The disputed meaning of the word "weapon" will serve as an example of how a translation can appear correct but still lead people toward the wrong conclusion.

Justin Davis

Assigned role: *How They Learn the Alien Language / Communication Methods*

Justin is responsible for describing how Louise and the research team build understanding over time. His planned contribution will cover basic concepts, repetition, visual signs, comparison of symbols, pattern finding, and feedback. He will connect those methods to familiar NLP ideas such as labeled examples, training data, and checking an interpretation against new evidence.

Brandon Matias

Assigned role: *NLP Approaches (rule-based, statistical, deep NLP)*

Brandon is assigned to define and compare rule-based, statistical, and deep NLP approaches. He will explain how each approach could help organize patterns or support translation, then discuss its limits when the available data is small or the intended meaning depends on culture and context. He will use careful comparisons to show what each approach might contribute without treating one method as a complete solution.

Samantha Mireles

Assigned role: *Real-World NLP Tools and Examples*

Samantha is responsible for connecting the movie's communication problems to real-world NLP tools. Her planned examples include machine translation, speech recognition, chatbots, sentiment analysis, and language models. She will explain what these tools can do, where context can still be missed, and why human review remains important when a misunderstanding could have serious effects.

Elijah Raines

Assigned role: *Reflection, final lesson, references, and final assembly*

Elijah is assigned to write the reflection and state the final lesson about communication, context, patience, cooperation, and human judgment. He is also responsible for organizing the references actually used by the team, checking transitions and citation formatting, and assembling the submitted sections into one consistent report and presentation without changing each member's assigned focus.

Collaboration Summary

The six roles follow the planned flow of the project from the movie overview through the final reflection. Each member is expected to contribute the assigned section and one clear movie example. After the sections are shared, the team will check that the ideas connect, and Elijah will combine the material and references for the final submission.